

Stressful Life Events Mediate the Relationship Between Adverse Childhood Experiences and Insomnia

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Introduction

Adverse childhood experiences (ACEs) have been associated with sleep disorders in adolescents and adults (Kajeepeeta et al., 2015). Researchers have emphasized the importance of identifying mediators of this relationship (Greenfield et al., 2011). One potential mediator is stressful life events experienced in adulthood. Stress proliferation theory proposes that early adverse experiences may increase the likelihood of experiencing these subsequent stressful life events (Pearlin, 2010), with the more recent stressful events serving as a mechanism that connects ACEs to increased risk for mental/physical health issues in adulthood. Because recent stressful life events are predictive of insomnia (Liu et al., 2019), we hypothesized that self-reported ACEs would be positively associated with insomnia severity and that recent stressful life events would serve as a mediator of this ACEs-insomnia relationship.

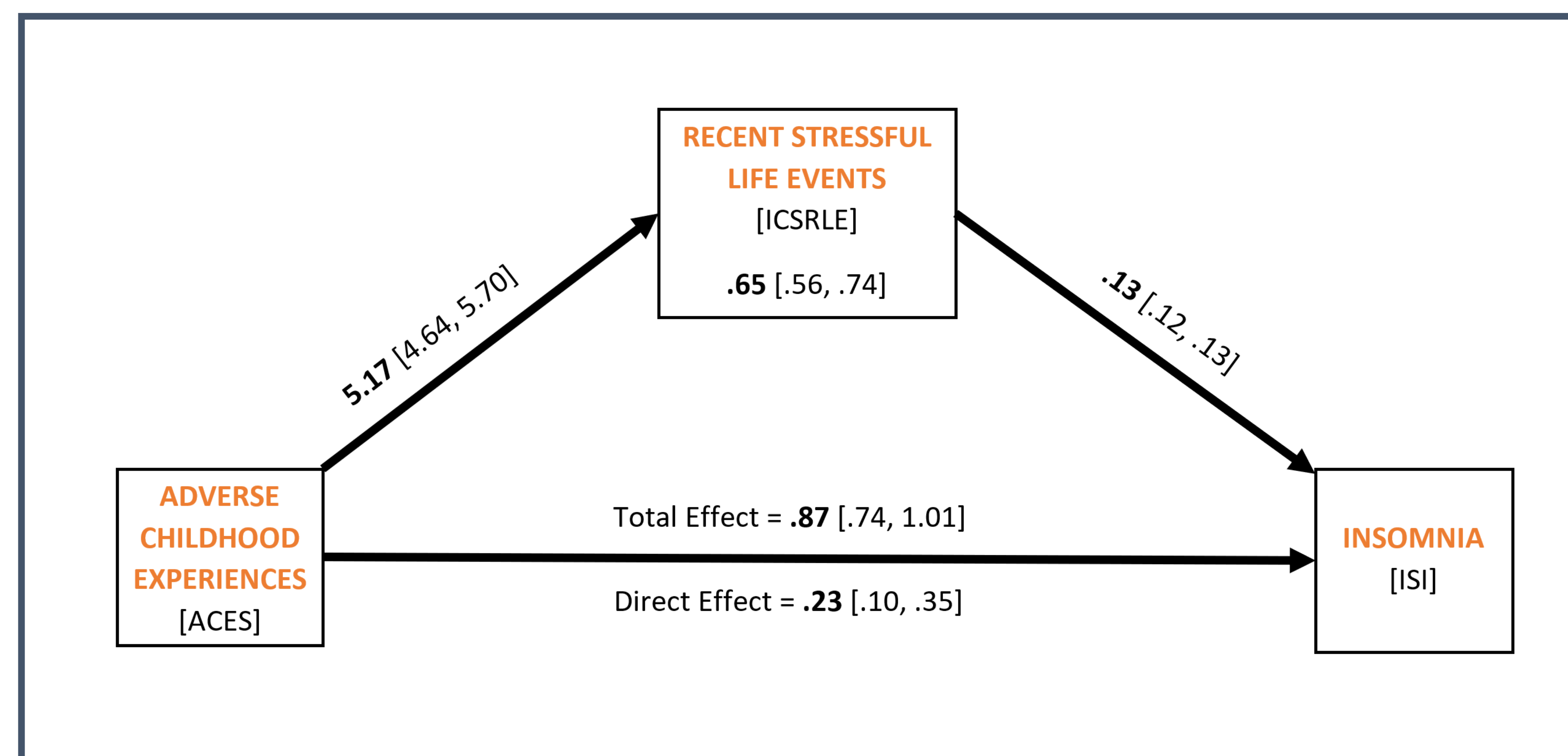
Methods

2108 undergraduates (75% female; 96% Hispanic; mean age = 20.7) completed an online anonymous survey that included measures of adverse childhood experiences (ACEs; Adverse Childhood Experiences Questionnaire), past-month stressful life events (Inventory of College Students' Recent Life Events) and past-month insomnia severity (Insomnia Severity Index). Pearson's correlation was used to measure the association between the variables. A mediation analysis using 5,000 bootstrapped samples was conducted using Hayes' PROCESS Macro V3.5. ACEs was examined as a predictor of insomnia severity, with stressful life events modeled as a mediator, controlling for age and gender.

- **Adverse Childhood Experiences Questionnaire (ACEs-Q).** A 10 –item survey assessing adverse childhood experiences.
- **Inventory of College Students' Recent Life Events (ICSRLE).** A 49-item survey assessing the frequency of students' experiences with hassles over the past month.
- **Insomnia Severity Index (ISI).** A 7-item survey assessing insomnia severity.

Results

Insomnia severity was positively associated with ACEs ($r(2070) = .29, p < .001$) and recent stressful life events ($r(2054) = .56, p < .001$). The ACEs-insomnia relationship was significantly mediated by stressful life events (unstandardized indirect effect = .65, SE = .05, 95% CI [.56, .74]). Overall, 32% of the variance in insomnia was accounted for by the model, $F(4,2046) = 238.0, p < .001$.



Conclusions

Our data demonstrate a statistically significant association between ACEs and insomnia severity in young adulthood, replicating prior findings. In support of stress proliferation theory, our confirmatory analysis demonstrated that stressful life events served as a mediator of the ACEs-insomnia relationship. These findings are in keeping with recent studies that found psychological distress (McPhie et al., 2014) and perceived stress (Cardoso et al., 2018) to mediate the relationship between ACEs and sleep disturbances. Efforts to reduce insomnia may benefit from understanding the role of childhood adversity and subsequent adult life events. For example, while it is not possible to alter college students' earlier childhood experiences, it may be possible to provide campus resources that help to reduce the number and/or intensity of common stressors such as social isolation and education/career decisions. We note some limitations to our study. Our sample was primarily Hispanic, and the generalizability of our results are therefore limited. However, this can be balanced against the relative paucity of data on this population, particularly in research on sleep health. Future research would benefit from longitudinal designs and the inclusion of objective measures of insomnia, such as polysomnography or actigraphy.

Adverse Childhood Experiences Questionnaire

- 1) Did you parent or other adult in the household **often...** swear at you, insult you, put you down or humiliate you?
or
Act in a way that made you afraid you might be physically hurt?
- 2) Did a parent or other adult in the household **often...** Push, grab, slap, or throw something at you?
or
Ever hit you so hard that you had marks or were injured?
- 3) Did a parent or other adult in the household **often...** Touch or fondle you or have you touch their body in a sexual way?
or
Try to or actually have oral, anal, or vaginal sex with you?
- 4) Did you **often** feel that...
No one in your family loved you or thought you were important or special?
or
Your family didn't look out for each other, feel close to each other, or support each other?
- 5) Did you **often** feel that...
You didn't have enough to eat, had to wear dirty clothes, and had no one to protect you?
or
Your parents were too drunk or high to take care of you or take you to the doctor if you needed it?
- 6) Were your parents ever separated or divorced?
- 7) Was you mother or stepmother:
Often pushed, grabbed, slapped, or had something thrown at her?
or
Sometimes or often kicked, bitten, hit with a fist, or hit with something hard?
or
Ever repeatedly hit over at least a few minutes or threatened with a gun or knife?
- 8) Did you live with anyone who was a problem drinker or alcoholic or who used street drugs?
- 9) Was a household member depressed or mentally ill or did a household member attempt suicide?
- 10) Did a household member go to prison?

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